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Forecast evaluation of The Model for Prediction Across Scales (MPAS) with a 70 km model top

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Abstract

The standard MPAS configuration uses a model-top height of 30 km, representing the atmosphere from the surface to the mid-stratosphere, with vertical wind damping applied above 22 km. While this sponge layer supports numerical stability, it can also suppress dynamical variability and influence model performance near the upper boundary. To assess the impacts of a higher model top on the stratospheric wind and temperature forecast, we configured MPAS extending to approximately 70 km, including the upper mesosphere. The vertical coordinate system below 30 km is kept identical to the standard configuration, with layer thickness gradually increasing above that level. Several dynamical damping parameters were revised for the sponge layer of 70 km model-top, and the Unified Gravity Wave Physics (UGWP) scheme was also tested. The European Centre for Medium-range Weather Forecasting (ECMWF) Reanalysis 5 (ERA5) model-level data were used for both initialization and forecast verification. The effects of the higher model top and UGWP were evaluated using cold-start forecasts on a quasi-uniform 120-km mesh, as well as a one-month cycling data-assimilation experiment with MPAS interfaced with the Joint Effort for Data assimilation Integration (MPAS-JEDI). We will also present the effort including ozone photochemistry parameterization (OPP) and using higher vertical resolution.

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